Data-driven Finite-horizon Optimal Control for Linear Time-varying Discrete-time Systems

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Abstract

This report presents a study of a data-driven approach for solving the finite-horizon optimal control problem for linear time-varying discrete-time (LTVDT) systems, based on the work of Pang, Bian, and Jiang [1]. We first investigate a model-based finite-horizon Policy Iteration (PI) method that iteratively solves the discrete-time Riccati equation without explicitly requiring the system matrices at every step. Building on this, we derive and implement a data-driven off-policy PI algorithm that finds approximate optimal controllers purely from input/state trajectory data, bypassing the need for any explicit system identification. The convergence of the algorithm is guaranteed under a mild rank condition on the collected data. Numerical results on a two-state linear time-varying system confirm that the value matrices converge monotonically to the optimal solution in only eight iterations, and the resulting controller substantially improves closed-loop state trajectories.

1 Introduction

Dynamic programming (DP), introduced by Bellman [2], laid the theoretical foundation for multi-stage optimal decision problems and has since driven significant progress in optimal control of dynamical systems. In practice, however, classical DP is impeded by two fundamental obstacles: the *Curse of Dimensionality*, arising from the exponential growth of the state space, and the *Curse of Modelling*, stemming from the requirement of an accurate mathematical model of the system. Reinforcement learning (RL) and adaptive/approximate dynamic programming (ADP) have emerged as promising paradigms to address these issues. Parametric function approximation techniques reduce the computational burden of value-function evaluation, while data-driven iterative algorithms circumvent explicit modelling by learning optimal controllers from plant interaction data.

A large body of existing work on model-free optimal control focuses on time-invariant, infinite-horizon problems, using Policy Iteration (PI) [4] or Value Iteration (VI). The finite-horizon, time-varying setting has received comparatively little attention. This is because, unlike the stationary infinite-horizon case, the optimal controller here is *non-stationary*, which introduces new challenges: the algebraic Riccati equation no longer applies, and the controller gains must be tracked backward in time over the entire horizon. Methods based on extremum seeking and dual-loop iteration have been proposed, but they typically require freshly generated data at each iteration, are restricted to single-input systems, or lack explicit convergence guarantees.

This report studies the approach proposed by Pang et al. [1], which addresses all three limitations simultaneously. The paper first establishes a model-based finite-horizon PI method for LTVDT systems as a finite-horizon counterpart to classical infinite-horizon PI, and then develops a data-driven off-policy PI algorithm that reuses a single batch of collected data across all iterations. Under a mild rank condition on the data, the algorithm is mathematically guaranteed to converge to the optimal solution.

The report is organized as follows. Section 2 formulates the optimal control problem and defines the relevant notation. Section 3 presents the model-based finite-horizon PI method along with its convergence result. Section 4 derives the off-policy, data-driven algorithm. Section 5 presents numerical results, and Section 6 concludes.

Notation

Throughout this report, $\mathbb{R}$ denotes the set of real numbers and $\mathbb{Z}$ the set of non-negative integers. $k \in \mathbb{Z}$ is the discrete time index. $X(k)$ is abbreviated as X_k. The Kronecker product is denoted by $\otimes$. For a matrix $A \in \mathbb{R}^{n\times m}$, $\text{vec}(A) = [a_1^T, a_2^T, \dots, a_m^T]^T$ where a_i is the i-th column. For a symmetric matrix $B \in \mathbb{R}^{m\times m}$, $\text{vecs}(B) \in \mathbb{R}^{\frac{1}{2}m(m+1)}$ stacks the upper triangular entries with off-diagonal entries doubled. For $v \in \mathbb{R}^n$, $\tilde{v} \in \mathbb{R}^{\frac{1}{2}n(n+1)}$ stacks all products $v_i v_j$, $i \le j$. $|\cdot|$ is the Euclidean norm for vectors and $\|\cdot\|$ is the induced matrix norm.

2 Problem Formulation and Preliminaries

Consider a linear time-varying discrete-time system:

$$x_{k+1} = A_k x_k + B_k u_k, \tag{1}$$

where $k \in \{k_0, k_0+1, \ldots, N-1\}$ is the discrete time instant, $x_k \in \mathbb{R}^n$ is the state vector, $u_k \in \mathbb{R}^m$ is the control input, and $A_k \in \mathbb{R}^{n\times n}$, $B_k \in \mathbb{R}^{n\times m}$ are time-dependent system matrices.

The goal is to find a sequence of control inputs $\{u_k\}_{k=k_0}^{N-1}$ that minimises the finite-horizon quadratic cost:

$$J(k_0, u) = x_N^T F x_N + \sum_{k=k_0}^{N-1} \big(x_k^T Q_k x_k + u_k^T R_k u_k\big), \quad (2)$$

where $F, Q_k \in \mathbb{R}^{n\times n}$ are symmetric positive semi-definite weighting matrices and $R_k \in \mathbb{R}^{m\times m}$ is symmetric positive definite.

Known-dynamics solution

When A_k and B_k are known for all k, this is the classical finite-horizon linear quadratic regulator (LQR). The optimal control law is a linear state-feedback:

$$u_k^* = -L_k^* x_k, \quad (3)$$

where the optimal gain matrices are:

$$L_k^* = (R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k, \quad (4)$$

and P_k satisfies the *discrete-time Riccati equation* (DRE), solved backward in time from the terminal condition $P_N = F$:

$$\begin{aligned} P_k &= Q_k + A_k^T P_{k+1} A_k \\ &- A_k^T P_{k+1} B_k (R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k. \end{aligned} \quad (5)$$

Motivation for a data-driven approach

When A_k and B_k are unknown, equation (5) cannot be applied directly. Furthermore, existing infinite-horizon data-driven methods are inapplicable here, because the algebraic Riccati equation used in the infinite-horizon setting differs structurally from the time-varying DRE (5). The key insight exploited in [1] is to rewrite the cost in terms of a *value matrix* associated with an arbitrary gain sequence, and then to express the Policy Evaluation step as a Lyapunov equation that can be solved using only observed data.

Assume an arbitrary gain sequence $\{L_k\}_{k=k_0}^{N-1}$ is applied, i.e., $u_k = -L_k x_k$. The closed-loop dynamics become:

$$x_{k+1} = A_{k,L} x_k, \quad A_{k,L} \triangleq A_k - B_k L_k. \quad (6)$$

The closed-loop state transition matrix Φ_L satisfies:

$$x_{k_1} = \Phi_L(k_1, k_2) x_{k_2}, \quad \Phi_L(k_1, k_2) = A_{k_1-1,L} \cdots A_{k_2,L}, \quad (7)$$

with $\Phi_L(k_1, k_1) = I$. The cost from time k under this gain sequence can then be written as:

$$J(k, u_L) = x_k^T V_{k,L}\, x_k, \quad (8)$$

where the symmetric positive semi-definite *value matrix* $V_{k,L}$ is:

$$\begin{aligned} V_{k,L} &= \Phi_L^T(N, k) F \Phi_L(N, k) \\ &+ \sum_{j=k}^{N-1} \Phi_L^T(j, k)(Q_j + L_j^T R_j L_j)\Phi_L(j, k), \end{aligned} \quad (9)$$

with $V_{N,L} = F$. Using (7) with $k_1 = k+1$, $k_2 = k$ in (9) yields the *Lyapunov equation*:

$$V_{k,L} = A_{k,L}^T V_{k+1,L} A_{k,L} + Q_k + L_k^T R_k L_k. \quad (10)$$

This equation is central to both the model-based and data-driven algorithms described in subsequent sections.

3 Finite-Horizon Policy Iteration

When the system matrices are known, the Lyapunov equation (10) can be solved iteratively to obtain the Riccati solution, without ever forming the Riccati recursion (5) directly. The procedure, stated as Theorem 1 in [1], alternates between two steps:

1. **Policy Evaluation:** Given the current gain sequence $\{L_{k,i}\}$, solve the Lyapunov equations backward from $V_{N,i} = F$:

   $$V_{k,i} = A_{k,i}^T V_{k+1,i} A_{k,i} + Q_k + L_{k,i}^T R_k L_{k,i}, \quad (11)$$

   where $A_{k,i} = A_k - B_k L_{k,i}$.

2. **Policy Improvement:** Update the gain matrices:

   $$L_{k,i+1} = (R_k + B_k^T V_{k+1,i} B_k)^{-1} B_k^T V_{k+1,i} A_k. \quad (12)$$

The convergence of this procedure is established by showing that the value matrix sequence is monotonically non-increasing and bounded below by the optimal Riccati solution P_k:

$$P_k \le V_{k,i+1} \le V_{k,i}, \quad \forall i \in \mathbb{Z}, \quad (13)$$

and that in the limit, $\lim_{i\to\infty} V_{k,i} = P_k$. The proof proceeds by induction: noting that at the terminal time $V_{N,i} = V_{N,i+1} = F$, and using the policy improvement equation (12) to show that the difference $V_{k,i} - V_{k,i+1}$ satisfies:

$$\begin{aligned} V_{k,i} - V_{k,i+1} &= A_{k,i+1}^T (V_{k+1,i} - V_{k+1,i+1}) A_{k,i+1} \\ &+ (L_{k,i} - L_{k,i+1})^T (R_k + B_k^T V_{k+1,i} B_k)(L_{k,i} - L_{k,i+1}), \end{aligned} \quad (14)$$

which is non-negative since $(R_k + B_k^T V_{k+1,i} B_k)$ is always positive definite. Since P_k corresponds to the value matrix under the *optimal* gain sequence, it is the lower bound. At convergence, the limit $V_{k,\infty}$ satisfies exactly the Riccati equation (5), and by uniqueness of the solution, $V_{k,\infty} = P_k$.

A key advantage of this formulation is that policy improvement step (12) has the same structure regardless of whether A_k and B_k are known, suggesting a path toward a data-driven implementation.

4 Data-driven Off-Policy Policy Iteration

When the system matrices are unknown, the Policy Evaluation step (11) cannot be solved directly. The data-driven algorithm replaces this step by a least-squares regression on input/state trajectory data, without identifying A_k or B_k explicitly.

4.1 Derivation of the data equation

Suppose the system is excited by a control sequence $u_{k,0,j} = -L_{k,0} x_j + w_{k,j}$, where $w_{k,j}$ is an exploration noise designed to ensure sufficient excitation. At the i-th policy iteration, the system equation (1) can be rewritten as:

$$x_{k+1} = A_{k,i} x_k + B_k (L_{k,i} x_k + u_{k,0}), \tag{15}$$

where $A_{k,i} = A_k - B_k L_{k,i}$. Substituting the Lyapunov equation (10) into the cost difference $x_{k+1}^T V_{k+1,i} x_{k+1} - x_k^T V_{k,i} x_k$ and rearranging yields the scalar data equation:

$$\psi_{k,i} = \phi_{k,i}\, \theta_{k,i}, \tag{16}$$

where $\psi_{k,i} = x_k^T (Q_k + L_{k,i}^T R_k L_{k,i}) x_k$ is computable from the current gain and state data, and $\phi_{k,i}$ is a row vector assembled from quadratic terms in x_k, x_{k+1}, and $u_{k,0}$. The unknown parameter vector $\theta_{k,i}$ contains the vectorised form of the matrices $V_{k,i}$, $B_k^T V_{k+1,i} A_k$, and $B_k^T V_{k+1,i} B_k$—the latter two being the quantities needed for the policy improvement step (12).

4.2 Batch data collection and least-squares solution

Since equation (16) is a single scalar equation with multiple unknowns, data from l independent trials must be stacked to form a determined system. Each trial j uses a different realization of the exploration noise $w_{k,j}$. Aggregating the data across all l trials and all time steps produces the matrix equation:

$$\Psi_i = \Phi_i\, \Theta_i, \tag{17}$$

where Θ_i stacks the unknown parameters for all time steps $k \in \{k_0, \ldots, N-1\}$, and Ψ_i, Φ_i are known data matrices assembled from the collected trajectories. If Φ_i has full column rank, the unique least-squares solution is:

$$\Theta_i = (\Phi_i^T \Phi_i)^{-1} \Phi_i^T \Psi_i. \tag{18}$$

From Θ_i, the gain matrix is updated via (12). The entire procedure reuses the *same* batch of collected data across all iterations, which is a notable practical advantage.

4.3 Convergence condition

The rank condition sufficient for convergence is:

$$\operatorname{rank}\big([\Gamma_{\tilde{x}_k}, \Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}]\big) = \tfrac{m(m+1)}{2} + mn + \tfrac{n(n+1)}{2}, \tag{19}$$

where $\Gamma_{\tilde{x}_k}$, $\Gamma_{xu_{k,0}}$, and $\Gamma_{\tilde{u}_{k,0}}$ are data matrices collecting quadratic state terms, cross state-input terms, and quadratic input terms, respectively, across all l trials. Condition (19) is a *persistence of excitation* requirement; it is satisfied when sufficiently rich exploration noise is used and l is large enough. When satisfied, Φ_i is guaranteed to have full column rank for all iterations i, and convergence of $V_{k,i} \to P_k$ and $L_{k,i} \to L_k^*$ follows by equivalence with the model-based algorithm of Section 3.

4.4 The complete algorithm

The full procedure is summarised in Algorithm 1.

Algorithm 1 Data-driven Off-policy Policy Iteration

Choose arbitrary $\{L_{k,0}\}_{k=k_0}^{N-1}$, threshold $\epsilon > 0$, $i \leftarrow 0$
Run system (1) l times; in run j, apply $u_{k,0,j} = -L_{k,0} x_j + w_{k,j}$
Collect and store all state/input trajectory data
repeat
    Compute Ψ_i and Φ_i from stored data and current $\{L_{k,i}\}$
    Solve $\Theta_i \leftarrow (\Phi_i^T \Phi_i)^{-1} \Phi_i^T \Psi_i$
    Extract $\{V_{k,i}\}$ and $\{B_k^T V_{k+1,i} A_k\}$ from Θ_i
    for $k = k_0$ to $N-1$ **do**
        $L_{k,i+1} \leftarrow (R_k + B_k^T V_{k+1,i} B_k)^{-1} B_k^T V_{k+1,i} A_k$
    end for
    $i \leftarrow i + 1$
until $\max_k \|V_{k,i} - V_{k,i-1}\| < \epsilon$
Output: $u_k = -L_{k,i} x_k$ as approximate optimal control

Remark on initialization. Unlike infinite-horizon PI methods, where stabilizing initial gains are required for convergence, the algorithm above is theoretically valid for any arbitrary initial gain sequence $\{L_{k,0}\}$. In practice, however, using a stabilizing initial sequence (when one is available) prevents states from growing excessively large during data collection [1].

Remark on direct adaptivity. No explicit system identification of (A_k, B_k) is performed at any stage. The method can therefore be classified as a *direct* adaptive control approach. Compared to indirect methods that first identify the system model and then compute a controller, the present approach needs to determine fewer unknowns when $m < n$, offering a computational advantage [1].

5 Numerical Example

We reproduce the numerical example from [1] to validate the algorithm. The system matrices are:

$$A_k = \begin{bmatrix} 1 & 0.0025k \\ -0.1\cos(0.3k) & 1 + (0.05)^{\frac{3}{2}}\sin(0.5k)\sqrt{k} \end{bmatrix}, \tag{20}$$

$$B_k = \begin{bmatrix} 1 \\ \dfrac{1}{(0.1k+2)(0.1k+3)} \end{bmatrix} \times 0.05. \tag{21}$$

The horizon is $N = 120$ with $k_0 = 0$. The cost weighting matrices are $Q_k = (0.04k + 2)I_2$, $R_k = 5 - 0.02k$, and $F = 50\, I_2$.

The initial gain matrix is set to $L_{k,0} = [0,\, 0]^T$ for all k. To satisfy the rank condition (19), the initial state elements are drawn independently and uniformly from $[-1, 1]$, and the exploration noise for the j-th trial is chosen as:

$$w_{k,j} = 2\sum_{s=1}^{500} \sin(\sigma_{s,j} \cdot k), \tag{22}$$

where each $\sigma_{s,j}$ is independently drawn from a uniform distribution over $[-500, 500]$. The convergence threshold is set to $\epsilon = 0.01$.

5.1 *Results: Model-based Policy Iteration*

To verify the finite-horizon PI method of Section 3, we first run the model-based algorithm using the known A_k and B_k. The PI controller drives both state components to zero, whereas the uncontrolled (zero-gain) system is unstable over the 120-step horizon. Figure **??** shows the comparison of state trajectories under the initial gain $L_{k,0}$ and under the PI-derived controller, alongside the resulting control input.

5.2 *Results: Off-policy Data-driven Algorithm*

Algorithm 1 is applied to the same system with the system matrices treated as *unknown*. The algorithm terminates after 8 iterations. Figure 1 shows the norm $\|V_{k,i} - P_k\|$ across all time steps for each iteration, split into two panels (iterations 1–4 and 4–8) for clarity. The value matrices converge monotonically toward the Riccati solution P_k, consistent with the theoretical guarantee. The convergence is rapid: the error drops by several orders of magnitude within the first four iterations.

Figure 2 shows the state trajectories under the data-driven approximate optimal controller, compared against the initial (zero-gain) controller. Both state components, starting from $x_0 = [-2,\, 1]^T$, are driven smoothly to zero under the approximate optimal controller, while the initial controller leads to growing, oscillatory trajectories. The corresponding control input is shown in Figure 3.

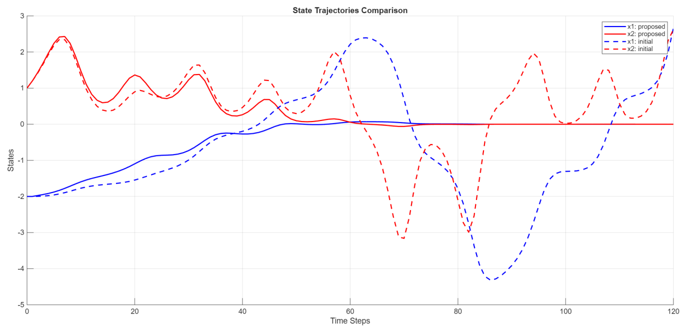

Figure 2: Comparison of state trajectories under the proposed data-driven approximate optimal controller (solid lines) vs. the initial zero-gain controller (dashed lines), for initial condition $x_0 = [-2,\, 1]^T$.

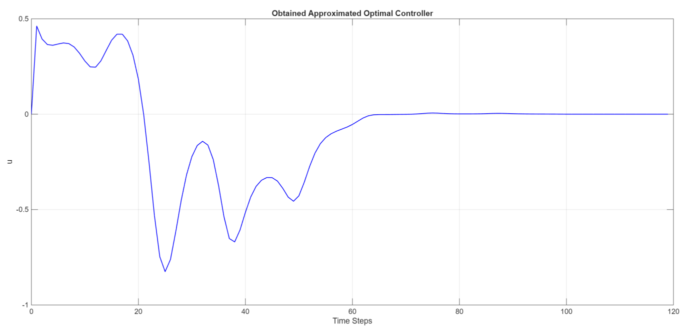

Figure 3: The approximate optimal control input $u_k = -L_{k,i^*}x_k$ produced by Algorithm 1 after convergence.

The results confirm that the off-policy data-driven algorithm closely replicates the performance of the model-based PI method, even without any knowledge of the system matrices A_k or B_k.

6 Conclusion

This report has studied a data-driven method for finite-horizon optimal control of linear time-varying discrete-time systems with unknown dynamics. The key contributions of the underlying paper [1] and the insights gained through this study are as follows.

First, a model-based finite-horizon Policy Iteration algorithm was presented that solves the discrete-time Riccati equation iteratively via a sequence of Lyapunov equations. The value matrix sequence is provably non-increasing and converges to the optimal Riccati solution.

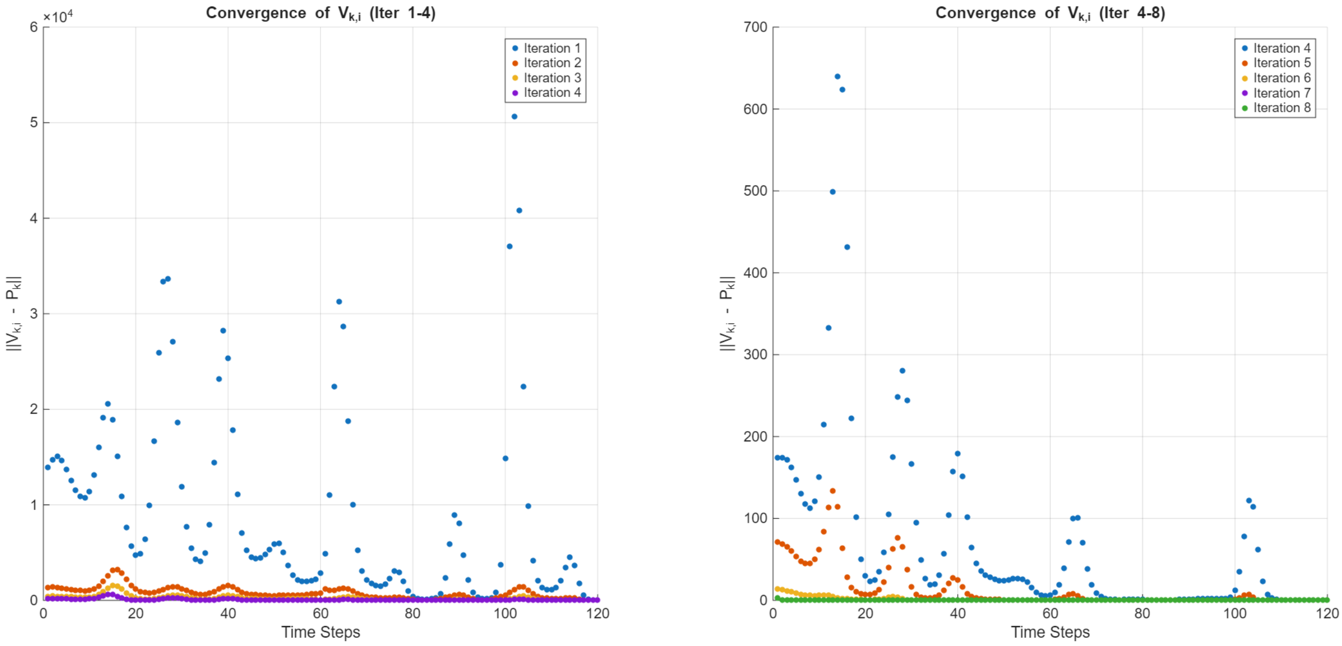

Figure 1: Convergence of $\|V_{k,i} - P_k\|$ across time steps for each iteration of Algorithm 1. *Left:* iterations 1–4; *Right:* iterations 4–8 (with inset showing fine-scale behavior near zero). The value matrices decrease monotonically to the optimal Riccati solution.

Second, leveraging the structure of the Lyapunov-based policy evaluation step, a data-driven off-policy PI algorithm was derived. By collecting input/state trajectory data from a single batch of experiments, and reusing this data across all subsequent iterations, the algorithm eliminates the need for both system identification and fresh data collection at each step. Convergence to the optimal controller is guaranteed as long as the rank condition (19) is satisfied—a persistence of excitation requirement that is easily enforced in practice through appropriate exploration noise design.

The numerical example demonstrated that the algorithm converges within 8 iterations on a two-state time-varying system, and that the resulting controller achieves near-identical performance to the model-based counterpart. This validates the approach as a practical and theoretically grounded solution to the curse of modelling in finite-horizon optimal control.

Possible directions for future work include extending the algorithm to nonlinear time-varying systems, incorporating robustness to measurement noise, and investigating the trade-off between the number of data collection trials l and the tightness of the rank condition.

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